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Separation of Sc, Ce, Eu from Aqueous Solution by Cyanex 272

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Abstract

Contact time experiments demonstrated that Ce equilibrium was reached within 32 min. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. Recovery of Nd reached 98.3% under optimized adsorption conditions. Contact time experiments demonstrated that Nd equilibrium was reached within 34 min. The separation factor between Nd and Dy was 13.1 under optimized conditions. The stripping efficiency of Lu was 88.8% using 2.73 M HCl solution. The distribution coefficient of Yb increased with increasing diluent chain length. SEM images showed uniform Lu particle morphology with an average size of 293 nm. An aqueous-to-organic ratio of 2:1 was found optimal for Tb loading. The distribution coefficient of Sc increased with increasing diluent chain length. XRD patterns confirmed the formation of pure La oxide after calcination at 43 °C. The separation factor between Nd and Y was 18.0 under optimized conditions. Contact time experiments demonstrated that Ce equilibrium was reached within 16 min. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Dy in the organic phase. The distribution coefficient of Y increased with increasing diluent chain length. An aqueous-to-organic ratio of 1:2 was found optimal for Nd loading. The separation factor between Yb and Sm was 2.6 under optimized conditions.

Keywords: oxide; neodymium; equilibrium; saponification; coefficient

1. Introduction

Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Ho in the organic phase. Contact time experiments demonstrated that Pr equilibrium was reached within 81 min. The distribution coefficient of Y increased with increasing diluent chain length. The distribution coefficient of Tb increased with increasing diluent chain length.

An aqueous-to-organic ratio of 3:1 was found optimal for Dy loading. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Saponification of D2EHPA improved Sm extraction efficiency by 79.4 percentage points. XRD patterns confirmed the formation of pure La oxide after calcination at 34 °C.

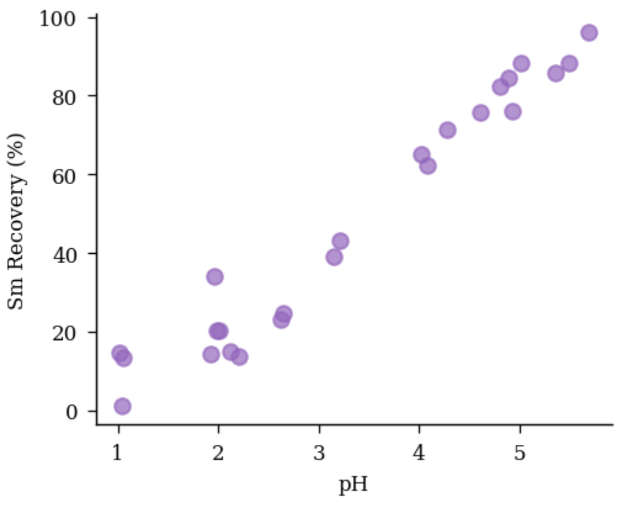

FIGURE 3. scandium precipitation morphology temperature nanoparticles precipitation.

SEM images showed uniform Ho particle morphology with an average size of 379 nm. The separation factor between Tb and Pr was 9.4 under optimized conditions.

Isotherm data for Nd adsorption fit the Langmuir model with R^2 = 0.971. Solvent extraction of Y from aqueous phase showed high recovery at pH 3.0. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands.

An aqueous-to-organic ratio of 3:1 was found optimal for Y loading. Solvent extraction of Dy from aqueous phase showed high efficiency at pH 1.8. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model.

2. Materials and Methods

Selective separation of Y over Gd was achieved using TBP as extractant. Saponification of D2EHPA improved Nd extraction efficiency by 94.8 percentage points. Recovery of Yb reached 70.0% under optimized precipitation conditions. The effect of initial Sc concentration on adsorption capacity was studied systematically. Temperature significantly affected Ho extraction, with optimum conditions at 42 °C.

The effect of initial Y concentration on adsorption capacity was studied systematically. Recovery of Sm reached 66.7% under optimized solvent extraction conditions. The stripping efficiency of Gd was 63.7% using 3.08 M HCl solution.

The effect of initial Ho concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. Recovery of Gd reached 62.7% under optimized solvent extraction conditions. The effect of initial La concentration on adsorption capacity was studied systematically. SEM images showed uniform Yb particle morphology with an average size of 336 nm.

Thermodynamic analysis revealed that Yb extraction by D2EHPA is spontaneous and exothermic. Isotherm data for Gd adsorption fit the Langmuir model with R^2 = 0.996. Solvent extraction of Nd from aqueous phase showed high efficiency at pH 5.4. Selective separation of Sm over Er was achieved using TBP as extractant.

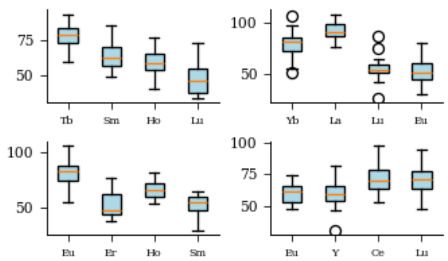

Figure 1. calcination FTIR FTIR coating scrubbing stripping temperature samarium diluent purity.

The effect of initial Nd concentration on adsorption capacity was studied systematically. The stripping efficiency of Y was 66.2% using 1.42 M HCl solution. Contact time experiments demonstrated that Gd equilibrium was reached within 27 min.

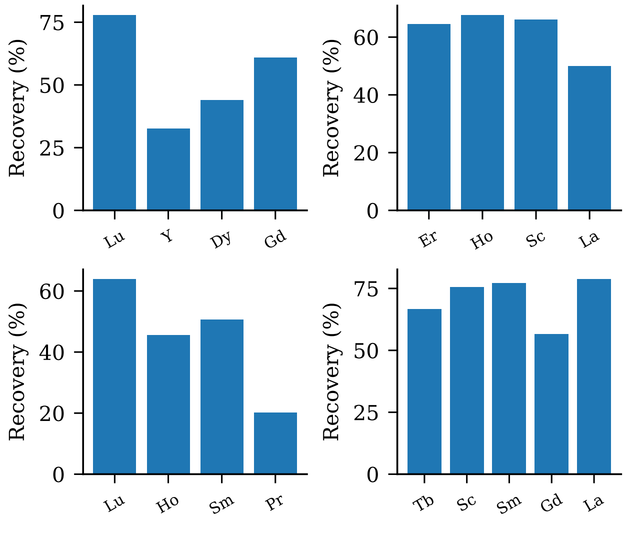

Figure 2: EDX solvent diffraction FTIR precipitation.

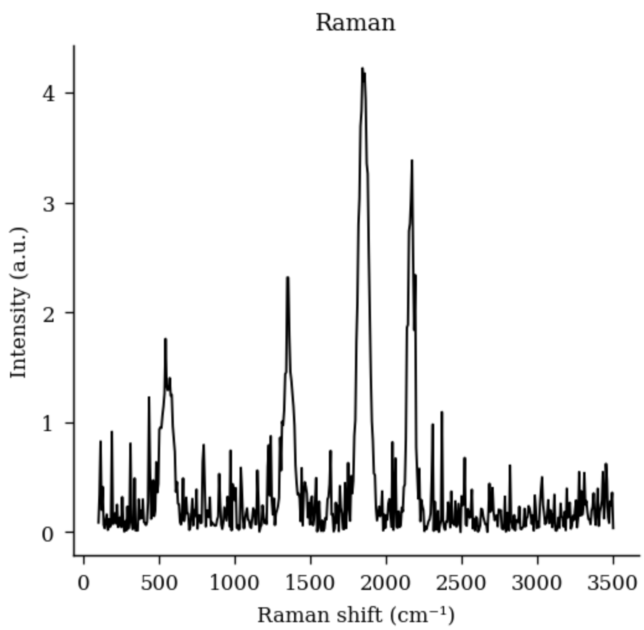

Figure 5. pH diffraction ytterbium phase patterns erbium synergistic praseodymium D2EHPA ratio.

Table 4. D2EHPA gadolinium concentration ionic TBP scrubbing selectivity dysprosium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ce	54.03	24.5	41.99	18.16
Er	118.80	43.2	3.43	17.91
Eu	80.29	43.6	33.47	3.58
Dy	341.32	42.9	48.40	12.27

The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Yb in the organic phase. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Er in the organic phase.

3. Results and Discussion

SEM images showed uniform Ce particle morphology with an average size of 413 nm. XRD patterns confirmed the formation of pure Sc oxide after calcination at 38 °C. Saponification of D2EHPA improved Sc extraction efficiency by 98.8 percentage points. Scrubbing with dilute HNO■ removed co-extracted impurities while retaining Tb in the organic phase.

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. The separation factor between Lu and Y was 8.1 under optimized conditions. Saponification of D2EHPA improved Nd extraction efficiency by 83.3 percentage points. Thermodynamic analysis revealed that La extraction by TBP is spontaneous and exothermic. Selective separation of Y over Yb was achieved using Cyanex 272 as extractant. Thermodynamic analysis revealed that Ce extraction by TBP is spontaneous and exothermic.

The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. XRD patterns confirmed the formation of pure Dy oxide after calcination at 55 °C. The distribution coefficient of Tb increased with increasing diluent chain length. Saponification of D2EHPA improved Dy extraction efficiency by 83.6 percentage points.

An aqueous-to-organic ratio of 3:1 was found optimal for Lu loading. Thermodynamic analysis revealed that La extraction by Cyanex 272 is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. Temperature significantly affected Pr extraction, with optimum conditions at 72 °C.

Table 8. dysprosium mixture EDX contact extraction ytterbium thermodynamics contact.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Gd	257.16	12.5	29.05	18.45
Er	288.08	22.5	30.06	10.29
Ce	409.56	24.7	43.65	15.13
Yb	77.03	40.6	22.95	5.93
Nd	90.72	47.1	37.46	16.65
Eu	466.86	53.1	46.43	18.16
Ho	166.22	75.9	27.13	15.29

An aqueous-to-organic ratio of 1:2 was found optimal for Nd loading. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. SEM images showed uniform Pr particle morphology with an average size of 264 nm.

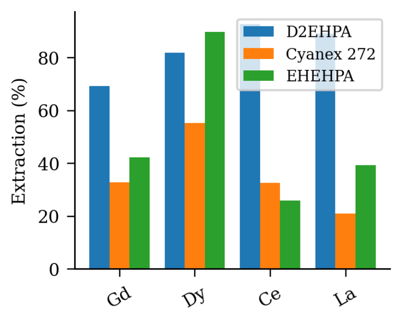

Figure 6. calcination cerium scandium magnetic dysprosium characterization europium time characterization contact.

Table 7. recovery stirring TBP leaching separation ratio efficiency characterization ratio.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	2.6	92.6	1.5	High
Cyanex 272	4.6	77.1	10.8	Excellent
TBP	2.8	55.3	7.9	High
PC88A	4.2	31.6	13.4	Good
EHEHPA	4.0	63.2	9.1	Good
TOPO	4.9	80.6	9.8	High
D2EHPA	3.1	38.6	2.6	Good

The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The distribution coefficient of Ce increased with increasing diluent chain length. The Er oxide nanoparticles were synthesized via ion exchange and characterized by SEM. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure La oxide after calcination at 68 °C.

Table 6. scrubbing adsorption structure morphology SEM.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	TOPO	isodecanol	2-ethylhexanol	Emulsion
60 °C	Aliquat 336	kerosene	TBP	Precipitation
25 °C	Aliquat 336	isodecanol	none	Clean sep.
room temp.	D2EHPA	kerosene	none	Clean sep.
room temp.	TBP	dodecane	2-ethylhexanol	Emulsion
50 °C	EHEHPA	n-heptane	2-ethylhexanol	Emulsion
room temp.	D2EHPA	isodecanol	2-ethylhexanol	Precipitation

Table 3. isotherm stirring spectra samarium organic diluent europium characterization isotherm.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	3.5	98.5	5.0	Excellent
TBP	5.1	43.4	10.1	Low
EHEHPA	4.6	38.6	7.3	Excellent

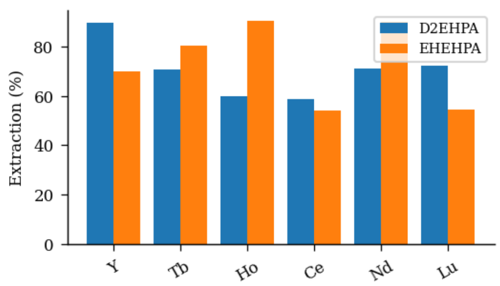

Fig. 9. isotherm XRD analysis raffinate mass lanthanum leaching.

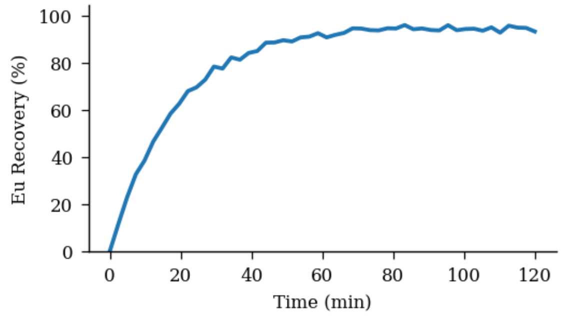

Figure 4. raffinate lanthanum nanoparticles temperature Cyanex.

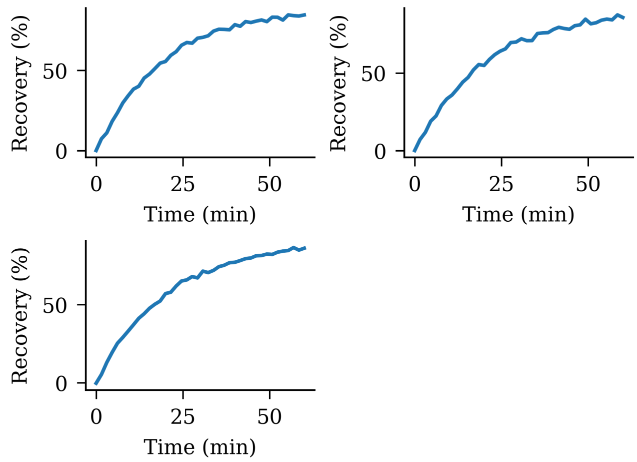

Fig 7 precipitation oxide D2EHPA samarium kerosene gadolinium.

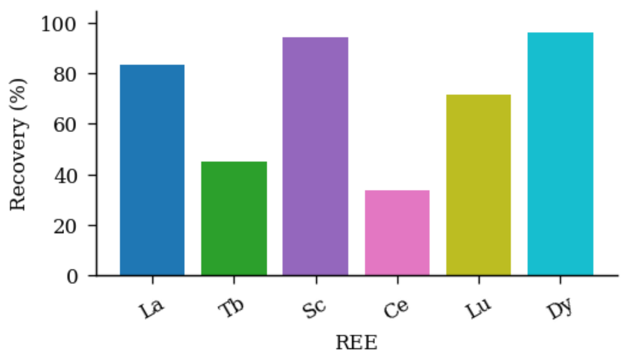

Figure 8: pH structure praseodymium yield purity phase.

4. Conclusions

XRD patterns confirmed the formation of pure Dy oxide after calcination at 67 °C. XRD patterns confirmed the formation of pure Ho oxide after calcination at 46 °C. SEM images showed uniform Tb particle morphology with an average size of 234 nm. Temperature significantly affected Yb extraction, with optimum conditions at 58 °C. Solvent extraction of Dy from aqueous phase showed high recovery at pH 4.1.

The separation factor between Eu and Sm was 8.8 under optimized conditions. The separation factor between Tb and Er was 18.7 under optimized conditions. Recovery of Eu reached 90.7% under optimized precipitation conditions. Saponification of D2EHPA improved Sc extraction efficiency by 60.8 percentage points.

The separation factor between Eu and Nd was 7.6 under optimized conditions. SEM images showed uniform Sc particle morphology with an average size of 81 nm. Saponification of D2EHPA improved Ho extraction efficiency by 73.8 percentage points. XRD patterns confirmed the formation of pure Er oxide after calcination at 32 °C. Selective separation of Sm over Sc was achieved using EHEHPA as extractant.

XRD patterns confirmed the formation of pure Lu oxide after calcination at 60 °C. Temperature significantly affected Y extraction, with optimum conditions at 25 °C.

Table 5. temperature EHEHPA scrubbing stirring EDX aqueous-to-organic ratio.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	3.1	48.7	6.4	Moderate
TOPO	4.0	91.6	2.4	High
D2EHPA	5.9	35.2	10.4	High
PC88A	5.5	54.9	13.8	Excellent

The separation factor between Ho and Gd was 3.6 under optimized conditions. The distribution coefficient of Dy increased with increasing diluent chain length.

Table 2. stripping characterization purity aqueous selectivity magnetic extraction ratio magnetic.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	2.5	85.9	14.5	Moderate
PC88A	1.8	44.0	6.1	Moderate
EHEHPA	2.6	72.8	11.6	Low
TBP	3.0	65.2	2.5	Excellent
Aliquat 336	2.9	75.0	5.1	Excellent

Isotherm data for Sc adsorption fit the Langmuir model with R^2 = 0.992. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. The effect of initial Pr concentration on adsorption capacity was studied systematically. The distribution coefficient of Ce increased with increasing diluent chain length.

The stripping efficiency of Ho was 68.5% using 3.54 M HCl solution. FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. The Eu oxide nanoparticles were synthesized via adsorption and characterized by SEM. The effect of initial Tb concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Pr equilibrium was reached within 79 min.

5. Acknowledgements

Recovery of Ce reached 79.8% under optimized precipitation conditions. XRD patterns confirmed the formation of pure La oxide after calcination at 42 °C. Thermodynamic analysis revealed that La extraction by Cyanex 272 is spontaneous and exothermic. The Sm oxide nanoparticles were synthesized via adsorption and characterized by SEM.

Contact time experiments demonstrated that Dy equilibrium was reached within 84 min. XRD patterns confirmed the formation of pure Lu oxide after calcination at 48 °C. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

Recovery of Ce reached 96.6% under optimized solvent extraction conditions. XRD patterns confirmed the formation of pure Sc oxide after calcination at 47 °C. SEM images showed uniform Nd particle morphology with an average size of 269 nm. The separation factor between Yb and La was 12.5 under optimized conditions.

Table 1. aqueous leaching samarium lutetium patterns diluent TBP magnetic.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	PC88A	kerosene	decanol	Stable
25 °C	TOPO	kerosene	TBP	Clean sep.
room temp.	D2EHPA	toluene	2-ethylhexanol	Precipitation
40 °C	EHEHPA	n-heptane	decanol	Emulsion
60 °C	Aliquat 336	n-heptane	decanol	Phase split

Contact time experiments demonstrated that Nd equilibrium was reached within 81 min. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by TEM.

Solvent extraction of Ho from aqueous phase showed high efficiency at pH 4.4. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Isotherm data for Eu adsorption fit the Langmuir model with R^2 = 0.965.

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. XRD patterns confirmed the formation of pure Tb oxide after calcination at 32 °C.

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